

Keisuke YANAGISAWA

Associate Professor, School of Computing, Institute of Science Tokyo



Current Status: Associate Professor

Address: 2-12-1 W8-87, Ookayama, Meguro-ku, Tokyo, 152-8550 Japan

Phone: +81-3-5734-3645 (Office)

Email: yanagisawa [at] comp.isct.ac.jp

Personal Website: <https://keisuke-yanagisawa.github.io>

Lab Website: <https://www.bi.comp.isct.ac.jp>

Date of Birth: 13 May 1991

Place of Birth: Tokyo, Japan

Citizenship: Japan

Education:

- 04/2016-03/2019 Department of Computer Science, School of Computing, Tokyo Institute of Technology, D.Eng.
- 04/2014-03/2016 Department of Computer Science, Graduate School of Information Science and Engineering, Tokyo Institute of Technology, M.Eng.
- 04/2010-03/2014 Department of Computer Science, Faculty of Engineering, Tokyo Institute of Technology, B.Eng.

Employment:

- 04/2026-present Associate Professor, Institute of Science Tokyo
- 10/2024-03/2026 Assistant Professor, Institute of Science Tokyo
- 09/2020-present Part-time Lecturer, Tokyo Tech Innovation
- 03/2020-09/2024 Assistant Professor, Tokyo Institute of Technology
- 11/2019-03/2020 Part-time Lecturer, Tokyo Institute of Technology
- 04/2019-03/2020 JSPS Research Fellow (PD), Japan Society for the Promotion of Science
- 04/2017-03/2019 JSPS Research Fellow (DC2), Japan Society for the Promotion of Science

Grants:

- FY2026-FY2029 “Development of protein structure-based methods to select an appropriate drug modality”, JSPS KAKENHI (grant-in-aid for Scientific Research (B)), 26K03025, PI, 18 330 000 yen
- FY2025-FY2029 “A virtual screening method using representative fragment-based reduced compound libraries”, JSPS KAKENHI (grant-in-aid for Scientific Research (B)), 25K03215, 18 850 000 yen
(PI: Yutaka Akiyama)
- FY2024-FY2026 “Improvement of protein-ligand docking scoring function with mixed-solvent molecular dynamics”, Kayamori Foundation of Informational Science Advancement Research Funding Program, PI, 1 000 000 yen
- FY2023-FY2025 “Development of large-scale virtual screening method for quantum/AI hybrid drug discovery”, NEDO Development of Quantum-Classical Hybrid Use-Case Technologies in Cyber-Physical Space,
224 587 000 yen (PI: Motokazu Iwasaki)
- FY2023-FY2025 “Designing cyclic peptide with target protein selection based on mixed-solvent molecular dynamics simulation”, JSPS KAKENHI (grant-in-aid for Scientific Research (B)), 23H03495/23K28185, PI, 18 590 000 yen
- FY2022-FY2024 “Development of chemical substructure-based virtual screening method for huge compound library”, JSPS KAKENHI (grant-in-aid for Scientific Research (B)), 22H03684/23K24939, 17 030 000 yen
(PI: Yutaka Akiyama)
- FY2020-FY2022 “Comprehensive prediction of cryptic binding sites by multi-task deep learning”, JSPS KAKENHI (grant-in-aid for young researchers), 20K19917, PI, 4 290 000 yen
- FY2020 “Computer-aided lead optimization with cosolvent molecular dynamics (CMD)”, Grant for young researcher, School of Computing, Tokyo Institute of Technology, PI, 496 000 yen
- FY2019 “Improvement of cosolvent MD which enables the systematic search of binding sites and the novel screening way of drug candidates”, JSPS KAKENHI (grant-in-aid for JSPS fellows), 19J00878, PI, 1 820 000 yen
- FY2017-FY2018 “Development of divide-and-conquer based docking method using common partial structures of hundreds of millions of compounds”, JSPS KAKENHI (grant-in-aid for JSPS fellows), 17J06897, PI, 2 100 000 yen

Awards:

- 03/2021 2019 Education Award of Excellence,
Tokyo Institute of Technology (as a member out of 17 members)
- 02/2020 Seiichi Tejima Doctoral Dissertation Award, Tokyo Institute of Technology.

- 12/2017 Grand Prize (Schrodinger K.K. Prize),
4th Computer-Aided Drug Discovery Contest.
- 07/2015 Student Encouragement Prize,
2nd Computer-Aided Drug Discovery Contest.
- 06/2015 2014 SIGBIO Best Student Presentation Award, IPSJ SIGBIO.
- 07/2014 Student Encouragement Prize,
1st Computer-Aided Drug Discovery Contest.
- 03/2014 2013 Tokyo Institute of Technology Academic Excellence Awards,
Tokyo Institute of Technology.
- 09/2009 Bronze Prize, 2009 High School Chemistry Grand Prix.

Current Research Interests:

Molecular Dynamics, Computational Biology, Bioinformatics, Structural Biology,
Protein-Ligand Docking, Protein-Ligand Interaction, Computational Drug Discovery,
Virtual Screening, Structure-based Virtual Screening, Machine Learning,
Quantum Annealing

Publications:

Peer-reviewed Journal Papers

1. Masayoshi Shimizu, Satoshi Yoneyama, **Keisuke Yanagisawa**, Yutaka Akiyama. “COFFEE-PRESC: A Fast Prescreening Method Using Compound Retrieval by Pairwise Positional Relationship of Representative Fragments”, *Journal of Chemical Information and Modeling*, 04/2026. (Epub ahead of print)
2. Jun Koseki, Chie Motono, **Keisuke Yanagisawa**, Genki Kudo, Ryunosuke Yoshino, Takatsugu Hirokawa, Kenichiro Imai. “CrypToth: Cryptic Pocket Detection through Mixed-Solvent Molecular Dynamics Simulations-Based Topological Data Analysis”, *Journal of Chemical Information and Modeling*, 65: 5567-5575, 06/2025.
3. Chie Motono, **Keisuke Yanagisawa**, Jun Koseki, Kenichiro Imai. “CrypTothML: An Integrated Mixed-Solvent Molecular Dynamics Simulation and Machine Learning Approach for Cryptic Site Prediction”, *International Journal of Molecular Sciences*, 26: 4710, 05/2025. (15 pages)
4. Jianan Li, **Keisuke Yanagisawa**, Yutaka Akiyama. “CycPeptMP: enhancing membrane permeability prediction of cyclic peptides with multi-level molecular features and data augmentation”, *Briefings in Bioinformatics*, 25: bbae417, 07/2024. (12 pages)
5. **Keisuke Yanagisawa**, Takuya Fujie, Kazuki Takabatake, Yutaka Akiyama. “QUBO Problem Formulation of Fragment-Based Protein-Ligand Flexible Docking”, *Entropy*, 26: 397, 04/2024. (15 pages)
6. Genki Kudo[†], **Keisuke Yanagisawa**[†], Ryunosuke Yoshino, Takatsugu Hirokawa. “AAp-MSMD: Amino Acid Preference Mapping on Protein-Protein Interaction Surfaces Using Mixed-Solvent Molecular Dynamics”, *Journal of Chemical Information and Modeling*, 63: 7768-7777, 12/2023. (†: Contributed equally)
7. Jianan Li, **Keisuke Yanagisawa**, Masatake Sugita, Takuya Fujie, Masahito Ohue, Yutaka Akiyama. “CycPeptMPDB: A Comprehensive Database of Membrane Permeability of Cyclic Peptides”, *Journal of Chemical Information and Modeling*, 63: 2240-2250, 03/2023.
8. Masatake Sugita, Takuya Fujie, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “Lipid composition is critical for accurate membrane permeability prediction of cyclic peptides by molecular dynamics simulations”, *Journal of Chemical Information and Modeling*, 62: 4549-4560, 09/2022.
9. **Keisuke Yanagisawa**, Rikuto Kubota, Yasushi Yoshikawa, Masahito Ohue, Yutaka Akiyama. “Effective protein-ligand docking strategy via fragment reuse and a proof-of-concept implementation”, *ACS Omega*, 7: 30265-30274, 08/2022.
10. **Keisuke Yanagisawa**, Ryunosuke Yoshino, Genki Kudo, Takatsugu Hirokawa. “Inverse Mixed-Solvent Molecular Dynamics for Visualization of the Residue Interaction Profile of Molecular Probes”, *International Journal of Molecular Sciences*, 23: 4749, 04/2022. (14 pages)

11. Kazuki Takabatake, **Keisuke Yanagisawa**, Yutaka Akiyama. “Solving Generalized Polyomino Puzzles Using the Ising Model”, *Entropy*, 24: 354, 02/2022. (21 pages)
12. Jianan Li, **Keisuke Yanagisawa**, Yasushi Yoshikawa, Masahito Ohue, Yutaka Akiyama. “Plasma protein binding prediction focusing on residue-level features and circularity of cyclic peptides by deep learning”, *Bioinformatics*, 38: 1110-1117, 02/2022.
13. **Keisuke Yanagisawa**. “Virtual Screening Methods with a Protein Tertiary Structure for Drug Discovery”, *JSBi Bioinformatics Review*, 2: 76-86, 10/2021. (in Japanese)
14. Kazuki Takabatake, Kazuki Izawa, Motohiro Akikawa, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “Improved Large-Scale Homology Search by Two-step Seed Search Using Multiple Reduced Amino Acid Alphabets”, *Genes*, 12, 1455, 09/2021. (12 pages)
15. Masatake Sugita, Satoshi Sugiyama, Takuya Fujie, Yasushi Yoshikawa, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “Large-Scale Membrane Permeability Prediction of Cyclic Peptides Crossing a Lipid Bilayer Based on Enhanced Sampling Molecular Dynamics Simulations”, *Journal of Chemical Information and Modeling*, 61, 3681–3695, 07/2021.
16. **Keisuke Yanagisawa**, Yoshitaka Moriwaki, Tohru Terada, Kentaro Shimizu. “EXPRORER: Rational cosolvent set construction method for cosolvent molecular dynamics using large-scale computation”, *Journal of Chemical Information and Modeling*, 61, 2744–2753, 06/2021.
17. Masahiro Mochizuki, Shogo D. Suzuki, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “QEX: Target-specific druglikeness filter enhances ligand-based virtual screening”, *Molecular Diversity*, 23, 11–18, 02/2019.
18. Takashi Tajimi, Naoki Wakui, **Keisuke Yanagisawa**, Yasushi Yoshikawa, Masahito Ohue, Yutaka Akiyama. “Computational prediction of plasma protein binding of cyclic peptides from small molecule experimental data using sparse modeling techniques”, *BMC Bioinformatics*, 19, 527, 12/2018. (14 pages)
19. **Keisuke Yanagisawa**, Shunta Komine, Rikuto Kubota, Masahito Ohue, Yutaka Akiyama. “Optimization of memory use of fragment extension-based protein-ligand docking with an original fast minimum cost flow algorithm”, *Computational Biology and Chemistry*, 74, 399–406, 06/2018.
20. Takanori Hayashi, Yuri Matsuzaki, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “MEGADOCK-Web: an integrated database of high-throughput structure-based protein-protein interaction predictions”, *BMC Bioinformatics*, 19, 62, 05/2018. (12 pages)
21. **Keisuke Yanagisawa**, Shunta Komine, Shogo D. Suzuki, Masahito Ohue, Takashi Ishida, Yutaka Akiyama. “Spresso: An ultrafast compound pre-screening method based on compound decomposition”, *Bioinformatics*, 33, 3836–3843, 03/2017.

22. Shuntaro Chiba, Takashi Ishida, Kazuyoshi Ikeda, Masahiro Mochizuki, Reiji Teramoto, Y-h. Taguchi, Mitsuo Iwadate, Hideaki Umeyama, Chandrasekaran Ramakrishnan, A. Mary Thangakani, D. Velmurugan, M. Michael Gromiha, Tatsuya Okuno, Koya Kato, Shintaro Minami, George Chikenji, Shogo D. Suzuki, **Keisuke Yanagisawa**, Woong-Hee Shin, Daisuke Kihara, Kazuki Z. Yamamoto, Yoshitaka Moriwaki, Nobuaki Yasuo, Ryunosuke Yoshino, Sergey Zozulya, Petro Borysko, Roman Stavniichuk, Teruki Honma, Takatsugu Hirokawa, Yutaka Akiyama, Masakazu Sekijima, “An iterative compound screening contest method for identifying target protein inhibitors using the tyrosine-protein kinase Yes”, *Scientific Reports*, 7, 12038, 09/2017. (13 pages)
23. Shuntaro Chiba, Kazuyoshi Ikeda, Takashi Ishida, M. Michael Gromiha, Y-h. Taguchi, Mitsuo Iwadate, Hideaki Umeyama, Kun-Yi Hsin, Hiroaki Kitano, Kazuki Yamamoto, Nobuyoshi Sugaya, Koya Kato, Tatsuya Okuno, George Chikenji, Masahiro Mochizuki, Nobuaki Yasuo, Ryunosuke Yoshino, **Keisuke Yanagisawa**, Tomohiro Ban, Reiji Teramoto, Chandrasekaran Ramakrishnan, A. Mary Thangakani, D. Velmurugan, Philip Prathipati, Junichi Ito, Yuko Tsuchiya, Kenji Mizuguchi, Teruki Honma, Takatsugu Hirokawa, Yutaka Akiyama, Masakazu Sekijima. “Identification of potential inhibitors based on compound proposal contest: Tyrosine-protein kinase Yes as a target”, *Scientific Reports*, 5, 17209, 12/2015. (13 pages)
24. **Keisuke Yanagisawa**, Takashi Ishida, Yutaka Akiyama. “Drug Clearance Pathway Prediction Based on Semi-supervised Learning”, *IPSJ Transactions on Bioinformatics*, 8, 21–27, 08/2015.

Peer-reviewed International Conference Papers

25. Kazuki Takabatake, Kazuki Izawa, Motohiro Akikawa, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “Improved Homology Search for Metagenomic Analysis by Two-Step Seed Search with Reduced Amino Acid Alphabets”, *The 10th International Conference on Bioinformatics and Biomedical Science (ICBBS2021)*, 10/2021.
26. Kazuya Isawa, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “Antisense oligonucleotide activity analysis based on opening and binding energies to targets”, In *Proceedings of the 27th International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA'21)*, 07/2021.
27. Masahito Ohue, Ryota Ii, **Keisuke Yanagisawa**, Yutaka Akiyama. “Molecular activity prediction using graph convolutional deep neural network considering distance on a molecular graph”, In *Proceedings of the 25th International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA'19)*, 07/2019.
28. Takashi Tajimi, Naoki Wakui, **Keisuke Yanagisawa**, Yasushi Yoshikawa, Masahito Ohue, Yutaka Akiyama. “Computational prediction of plasma protein binding of cyclic peptides from small molecule experimental data using sparse modeling techniques”, *The 29th International Conference on Genome Informatics (GIW 2018)*, 12/2018.
29. **Keisuke Yanagisawa**, Shunta Komine, Rikuto Kubota, Masahito Ohue, Yutaka Akiyama. “Optimization of memory use of fragment extension-based protein-ligand docking with an original fast minimum cost flow algorithm”, *The 16th Asia Pacific Bioinformatics Conference (APBC2018)*, 01/2018.
30. Takanori Hayashi, Yuri Matsuzaki, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “MEGADOCK-Web: an integrated database of high-throughput structure-based protein-protein interaction predictions”, *The 16th Asia Pacific Bioinformatics Conference (APBC2018)*, 01/2018.
31. **Keisuke Yanagisawa**, Shunta Komine, Shogo D. Suzuki, Masahito Ohue, Takashi Ishida, Yutaka Akiyama. “ESPRESSO: An ultrafast compound pre-screening method based on compound decomposition”, *The 27th International Conference on Genome Informatics (GIW 2016)*, 10/2016.

International Conferences (without peer review)

32. Chie Motono, **Keisuke Yanagisawa**, Jun Koseki, Kenichiro Imai, “CrypTothML: Cryptic Site Prediction using Mixed-Solvent Molecular Dynamics Simulation and Machine Learning”, *The international Chemical Congress of Pacific Basin Societies (Pacifichem) 2025*, 12/2025. (Poster presentation)
33. **Keisuke Yanagisawa**, Ryunosuke Yoshino, Genki Kudo, Takatsugu Hirokawa, “Quantitative Evaluation of Protein-Ligand Substructure Interaction with Inverse Mixed-Solvent Molecular Dynamics Simulation”, *Asia Hub for e-Drug Discovery 2025 (AHeDD2025)*, 09/2025. (Poster presentation)
34. Masatake Sugita, Yudai Noso, Jianan Li, Takuya Fujie, **Keisuke Yanagisawa**, Yutaka Akiyama, “Protocol for Membrane Permeability Prediction of Cyclic Peptides by Combining Molecular Dynamics Simulations and Machine Learning”, *Asia Hub for e-Drug Discovery 2025 (AHeDD2025)*, 09/2025. (Poster presentation)
35. Kaho Akaki, **Keisuke Yanagisawa**, Yutaka Akiyama, “Enhancing virtual screening accuracy by refining docking calculation scoring with mixed-solvent molecular dynamics”, *Asia Hub for e-Drug Discovery 2025 (AHeDD2025)*, 09/2025. (Poster presentation)
36. Masahiro Shimizu, Masatake Sugita, **Keisuke Yanagisawa**, Yutaka Akiyama, “An Automatic Iterative Refinement Protocol for Restraint Parameters in REUS Molecular Dynamics”, *Asia Hub for e-Drug Discovery 2025 (AHeDD2025)*, 09/2025. (Poster presentation)
37. Masayoshi Shimizu, Satoshi Yoneyama, **Keisuke Yanagisawa**, Yutaka Akiyama, “Development of a fast pre-screening method using compound retrieval by fragment pose pairs”, *Asia Hub for e-Drug Discovery 2025 (AHeDD2025)*, 09/2025. (Poster presentation)
38. **Keisuke Yanagisawa**, Takuya Fujie, Kazuki Takabatake, Yutaka Akiyama, “FraSCO-VS: Fragment-based drug virtual screening by combinatorial optimization with quantum annealer”, *Asia Hub for e-Drug Discovery 2025 (AHeDD2025)*, 09/2025. (Oral presentation)
39. **Keisuke Yanagisawa**, Ryunosuke Yoshino, Genki Kudo, Takatsugu Hirokawa. “Quantitative Evaluation of Protein-Compound Substructure Interaction with Inverse Mixed-Solvent Molecular Dynamics Simulation”, *21st IUPAB and 62nd BSJ joint congress 2024*, 06/2024. (Poster presentation)
40. **Keisuke Yanagisawa**, Yoshitaka Moriwaki, Tohru Terada, Kentaro Shimizu. “Systematic construction of the cosolvents sets for cosolvent MD (CMD) with the large-scale simulation”, *AHeDD2019/IPAB2019 Joint Symposium*, 11/2019. (Oral presentation)
41. Masahito Ohue, Takanori Hayashi, Yuri Matsuzaki, **Keisuke Yanagisawa**, Yutaka Akiyama. “Megadock-Web: An Integrated Database of High-Throughput Structure-Based Protein-Protein Interaction Predictions”, *Biophysical Society 63rd Annual Meeting*, 2792-Pos, 03/2019. (Poster presentation)

42. **Keisuke Yanagisawa**, Shunta Komine, Shogo D. Suzuki, Masahito Ohue, Takashi Ishida, Yutaka Akiyama. “Spresso: An ultrafast compound pre-screening method based on compound fragmentation”, *Biophysical Society 62nd Annual Meeting*, 02/2018. (Poster presentation)
43. Rikuto Kubota, **Keisuke Yanagisawa**, Masahito Ohue, Yutaka Akiyama. “Toward efficient protein-ligand docking for virtual screening by reuse of fragments”, *The 16th Asia Pacific Bioinformatics Conference (APBC2018)*, Poster C5, 01/2018. (Poster presentation)
44. **Keisuke Yanagisawa**, Shunta Komine, Masahito Ohue, Takashi Ishida, Yutaka Akiyama. “Fast pre-filtering for virtual screening based on compound fragmentation”, *3rd IIT Madras–Tokyo Tech Joint Symposium on Algorithms and Applications of Bioinformatics*, P34, 11/2015. (Poster presentation)